Neural Network Zoo: The CNN Cheetah

A03_TeamSolo_JavonDarby_ITAI2376
ITAI 2376 – Neural Network Zoo Assignment

Introduction to Neural Networks

It is made of small units called neurons that work together to process information and make predictions.

A neuron:

- Takes inputs
- Multiplies by weights
- Adds a bias
- Passes through an activation function
- Produces an output

Neurons are organized into layers:

- **Input Layer** – receives data
- **Hidden Layers** – processes data
- **Output Layer** – gives the result
- When many layers are used, this is called **Deep Learning**.

The Neural Network Zoo Concept

In this activity, each type of neural network is represented as an **animal in a zoo**.

Each animal's traits match how that neural network works.

My neural network animal is:

🐆 **CNN Cheetah (Convolutional Neural Network)**

Why a Cheetah Represents a CNN

- A cheetah has:
- Incredible vision
- Ability to notice small details
- Focus on patterns in its environment
- Fast decision-making ability

A **CNN** works the same way with images.

It scans images, detects patterns, and quickly classifies what it sees.

Structure of a Convolutional Neural Network

- A CNN is made of special layers:
- **Convolution Layer** – finds features like edges and shapes
- **ReLU Activation** – adds non-linearity
- **Pooling Layer** – reduces size but keeps important data
- **Fully Connected Layer** – makes the final decision

Flow of a CNN:

Input Image → Convolution → ReLU → Pooling → Fully Connected → Output

How the CNN “Sees” Like a Cheetah

Cheetah Behavior	CNN Behavior
Spots movement in grass	Detects edges in images
Focuses on patterns	Finds shapes and textures
Ignores background noise	Pooling removes extra data
Makes quick decisions	Classifies image fast

Real World Applications of CNN

CNNs are used in many real-world technologies:

- Face recognition (phones, security)
- Self-driving cars
- Medical image detection (cancer scans)
- Security cameras
- Object detection (cars, people, animals)
- Social media photo tagging

CNNs are the foundation of **computer vision AI**.

Similarities and Differences with Other Networks

Similarities:

- All neural networks use layers and neurons
- All learn from data
- All adjust weights to improve accuracy

Differences:

- **CNN** → Best for images and spatial patterns
- **RNN/LSTM** → Best for sequences, memory, text, and speech

CNN focuses on **vision**.
RNN/LSTM focuses on **time and memory**.

Reflection and Understanding

Through this activity, I learned how Convolutional Neural Networks specialize in visual pattern recognition.

Comparing CNNs to a cheetah helped me understand how the network focuses on important features in an image while ignoring unnecessary information.

I also understood how CNNs differ from other neural networks like RNNs and LSTMs, which are better suited for sequence-based data.

This zoo concept made it easier to connect animal traits to neural network functions and deepen my understanding of deep learning.

Thank You

Welcome to the Neural Network Zoo!
The CNN Cheetah is one of many neural network animals that help power modern AI systems.

Thank you for visiting the zoo.